



Recent applications of quantitative mass spectrometry in biopharmaceutical process development and manufacturing

Xuanwen Li

Analytical Research and Development, MRL, Merck & Co., Inc., 126 E. Lincoln Avenue, Rahway, NJ 07065, USA

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ABSTRACT

Biopharmaceutical products have seen rapid growth over the past few decades and continue to dominate the global pharmaceutical market. Aligning with the quality by design (QbD) framework and realization, recent advances in liquid chromatography-mass spectrometry (LC-MS) instrumentation and related techniques have enhanced biopharmaceutical characterization capabilities and have supported an increased development of biopharmaceutical products. Beyond its routine qualitative characterization, the quantitative feature of LC-MS has unique applications in biopharmaceutical process development and manufacturing. This review describes the recent applications and implications of the advancement of quantitative MS methods in biopharmaceutical process development, and characterization of biopharmaceutical product, product-related variants, and process-related impurities. We also provide insights on the emerging applications of quantitative MS in the lifecycle of biopharmaceutical product development including quality control in the Good Manufacturing Practice (GMP) environment and process analytical technology (PAT) practices during process development and manufacturing. Through collaboration with instrument and software vendors and regulatory agencies, we envision broader adoption of phase-appropriate quantitative MS-based methods for the analysis of biopharmaceutical products, which in turn has the potential to enable manufacture of higher quality products for patients.

1. Introduction

Quality by Design (QbD) is essential for the development of quality pharmaceutical products [1,2]. The International Council for Harmonisation (ICH) Q8 guidelines define quality as the suitability of either a drug substance or drug product for its intended use. The quality includes such attributes as the identity, strength, and purity. QbD enables a systematic approach for pharmaceutical development by beginning with predefined objectives with an emphasis on product and process understanding and process control, along with sound science and risk assessment [1,2]. Biopharmaceutical products produced by living cells often present a higher level of complexity than most small molecule pharmaceuticals in terms of manufacturing process or product structure [3]. Based on QbD concepts, manufacturing processes for biopharmaceuticals must be designed to produce products that have consistent quality attributes, that are linked to the identity, strength and purity of the product, and may ultimately relate to the safety and efficacy of the product [4]. The concept of QbD applied to analytical method development is known as analytical quality by design (AQbD) [5–7]. Implementation of AQbD depends on the target measurement for

understanding the product quality attributes (PQAs) and analytical target profile (ATP), followed by selection of suitable analytical techniques, risk assessments for various methods, method development, method robustness evaluation, method phase-appropriate validation, and life-cycle management [5–7].

While the product profile and degradation pathways for monoclonal antibodies (mAbs), a traditional biopharmaceutical, are relatively well-understood [8–11], the product and process understanding of newer modalities or advanced biopharmaceuticals, such as complex fusion proteins, antibody-drug conjugates (ADCs), certain vaccines, and cell and gene therapies (CGTs), is less developed [12–14]. A list of different biopharmaceutical modalities and their representative protein-based attributes are listed in Table 1. Due to the intricate mechanism of action (MOA) exhibited by advanced biopharmaceuticals, although certain critical quality attributes (CQAs) can be discerned to define some aspects of a product's safety and efficacy, numerous critical attributes of the product will inevitably remain largely unknown or unmeasurable [13,15]. Regulatory agencies recognize that identifying product-specific and clinically relevant CQAs can be challenging [1,2]. Accordingly, it is encouraged to use a systematic approach to define CQAs for

E-mail address: xuanwen.li@merck.com.

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Table 1

List of different biopharmaceutical modalities and their representative protein-based attributes.

Representative modalities*		Representative protein-based attributes
Therapeutic proteins	mAb, bispecific mAb, multi-specific mAb, enzymes, fusion proteins, conjugates (e.g., antibody-drug conjugate)	protein modifications (e.g., oxidation, deamidation, glycosylation), sequence variants, drug-to-antibody ratio, host cell proteins
Cell and gene therapies	viral vectors, CAR-T, CAR-NK, blood products	protein modifications, capsid stoichiometry, cell surface protein expression, host cell proteins
Vaccines	recombinant proteins, virus-like particles, conjugates (e.g., protein-polysaccharide conjugate), virus/viral vector	protein modifications, virus structure protein expression, capsid stoichiometry, host cell proteins

* Nucleic acid-based biopharmaceuticals are out of the scope of this review

biopharmaceuticals. The first step includes identifying potential CQAs based on the potential or known impact to efficacy and safety for each biopharmaceutical product. Based on these potential CQAs, analytical assays are developed to measure those CQAs, directly or indirectly. The accumulated knowledge gathered during the product development cycle forms a scientific basis for establishing meaningful acceptance criteria. Along with a systematic approach to correlate CQAs with product quality and clinical outcomes, the biological and clinical relevance of each candidate CQA can be established [13]. To support this highly systematic and iterative process, analytical methods capable of measuring molecule or site-specific attributes are the key part of the QbD paradigm.

Multiple well-established analytical assays, including polymerase chain reaction (PCR), sodium dodecyl-sulfate polyacrylamide gel electrophoresis (SDS-PAGE), high performance liquid chromatography (HPLC) and capillary electrophoresis (CE) based methods, immunoassays including Western blotting, enzyme-linked immunoassay (ELISA), and immunochemistry, biophysical and cell-based assays are currently used for biopharmaceuticals characterization throughout process development and manufacturing release [3,8,16,17]. Several conventional analytical assays are used to characterize the target molecules, individual biopharmaceutical components, or process residues at the intact molecule level. Consequently, the purity or impurity of the molecule is determined by the chosen method and not by the specific product quality attribute. If the release or stability data obtained from these assays fall beyond the specified limits, additional orthogonal attribute assays such as LC-MS assays can be employed to help identify the root cause [8,17]. With regulatory requirements becoming increasingly stringent, the pharmaceutical industry is under pressure to provide more comprehensive information on biopharmaceutical products while also accelerating development timelines. Compared to traditional biochemical assays, MS-based methods have demonstrated the ability to overcome some of the limitations of traditional assays by enabling the quantification of multiple amino acid site-specific product quality attributes with high specificity and sensitivity. This facilitates enhanced product and process understanding and supports the implementation of risk-based approaches to accelerate product development [8,17]. Achieving the quality targets is a high priority for biopharmaceutical manufacturers, who are facing increased competitive pressure from both innovator and biosimilar companies, where faster speed to market, higher product quality, and lower manufacturing costs are viewed as critical considerations to securing and retaining the market position of biopharmaceutical products [3].

Biological MS has become a powerful technology over the past 2–3 decades and has enabled both biomedical research and biopharmaceutical development [18–21]. Increased use of LC-MS for biopharmaceutical development, from discovery to development and manufacturing

phases, can be indirectly measured as a function of the increased number of articles published including terms relating to “mass spectrometry” and “biotherapeutics” as shown in Fig. 1. Across each of the 3 stages of biopharmaceutical development (Fig. 2A), the representative applications of LC-MS supporting cell line development, upstream cell culture, downstream purification and formulation development are summarized in Fig. 2B. This review primarily centers around exploring the applications of quantitative LC-MS in later-stage development and manufacturing. MS in nature is merely a qualitative analytical technique as variable ionization efficiencies of different molecules [19–21]. Quantitative data from MS often comes in two forms, the absolute amount of the target in a sample with isotopically labelled internal standards or the relative change of the same target between two different conditions [19–21]. Some of the quantification limitations without isotopically labelled internal standards can be minimized by normalizing between runs with spiked-in calibrants using abundant nonchanging peptides or proteins between runs. In most applications for biopharmaceutical process development, a rough and quick quantification as provided by relative quantification methods will be sufficient to guide subsequent follow-up process decision [8], while the full risk assessment will benefit from absolute quantification, especially for process-related residuals.

This review focuses on quantitative LC-MS approaches for protein analysis during biopharmaceutical process development for characterization and quality control of various biopharmaceutical products, product-related variants, and process-related impurities. The quantitative MS technologies with a line-of-sight to their implementation in manufacturing and GMP spaces are emphasized in this review. Other qualitative MS applications for extended biopharmaceutical characterization, such as top-down MS [22,23], high-order structure elucidation and native MS approaches [24–26], are out of scope of this review.

2. Review of current quantitative MS method

Quantitative MS technology has been well established for proteomics and bioanalysis of large molecules from various biological matrices [27, 28]. The properties of the list of the potential analytical methods to monitor product and process-related impurities were summarized in Table 2. Since some biopharmaceuticals, such as mAbs or fusion proteins, are generally well purified, intact mass and subunit mass-based high-resolution LC-MS methods allow for the quantification of intact proteins with little or no sample preparation (Table 2). MS can be used as a high throughput assay for monitoring the identity, purity, and protein modifications such as glycosylation and glycation of protein therapeutics (Table 1). For intact mass or subunit mass analysis by LC-MS, the mass spectrum of large molecules, such as mAbs (about 150 kDa), can be deconvoluted using dedicated software. Various charge states are converted into the corresponding and much simplified spectrum, which is subsequently used as the basis for quantitation assuming negligible difference of ionization efficiency among different molecules. While being straightforward and relatively easy to perform, the intact/subunit LC-MS methods have a few challenges (Table 2). First, the method in general is limited in its ability to monitor PQAs with small mass changes, such as deamidation or isomerization, or to generate site-specific information when multiple modification sites are located within the same subunit fragments. Second, sensitivity easily becomes an issue with increasing molecular weight, because of the distribution of the detection signal over multiple charged species, each with their own isotope distribution. The interpretation can be further complicated if the sample is a mixture of proteins due to the overlap of different protein charge envelopes, or quantification of low abundance protein variants in a sample with one dominant protein. The intact/subunit MS method may have challenges to quantify low-level PQA (e.g., <1%). To address the sensitivity issue, the quantification directly using the most abundant charge states of a protein can be considered as an alternative approach, which has showed similar selectivity, accuracy and precision as the

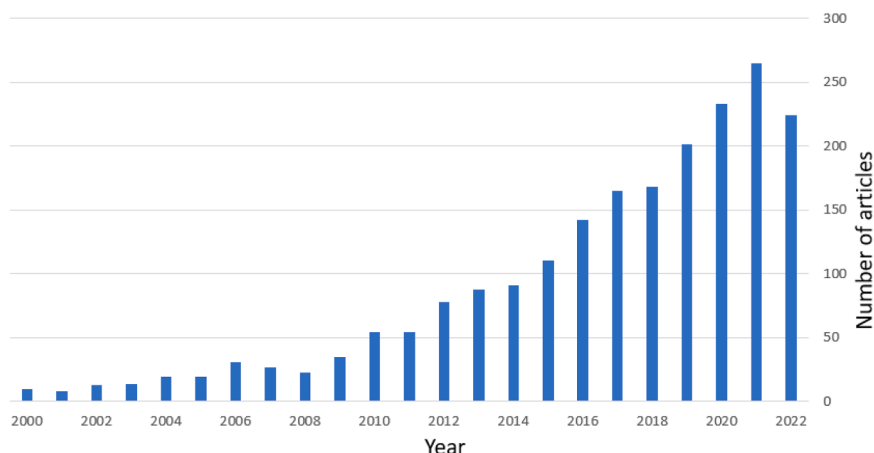


Fig. 1. Evolution of number of yearly publications with ("mass spectrometry"[All Fields] or "LC-MS"[All Fields]) AND ("biotherapeutics"[All Fields] OR "biologics"[All Fields] or "biopharmaceutical"[All Fields]) in Pubmed (update by Dec2022).

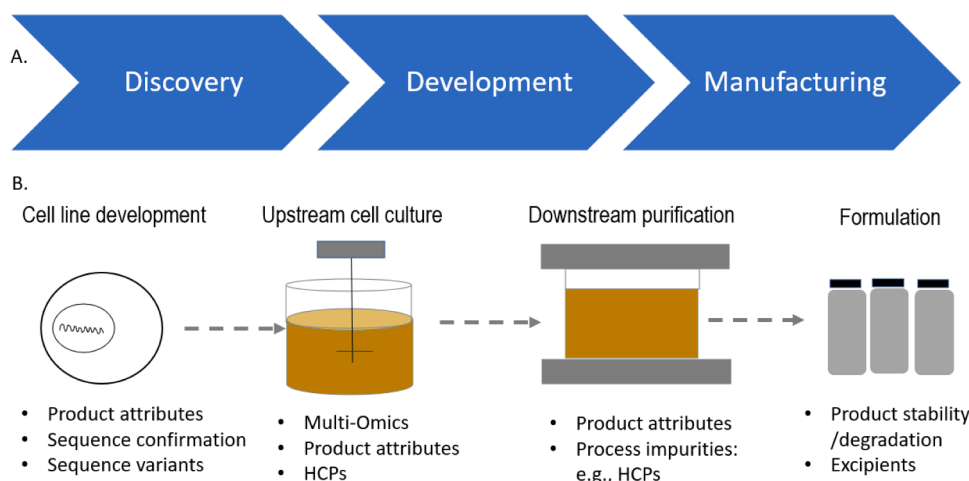


Fig. 2. Representative applications of LC-MS in various stages of biopharmaceutical development. A. different stages of biopharmaceutical development; B. representative applications of LC-MS supporting cell line development, upstream cell culture, downstream purification and formulation development across each of the 3 stages of biopharmaceutical development.

deconvolution approach in samples with biological matrix [29]. The intact/subunit mass analysis can be used for rapid verification of identity and characterization of sequence variants and protein modifications for both innovative and biosimilar products at different stages of the development [30–33]. Most recently, an intact mass analysis based intact multi-attribute method (iMAM) was developed. The authors evaluated the reproducibility of quantitative measurements of the relative abundance of protein modifications using the m/z values of the top 6 charge states. They found that the quantification variability was within 20% relative standard deviation (RSD) for most of protein modifications in the intact mass analysis-based iMAM method [34].

Bottom-up based peptide analysis by high- or low-resolution MS is still the most widely used approach for quantitative MS analysis of protein abundance or their modification levels (Table 2). With the bottom-up approach, proteins are enzymatically digested with trypsin or endoproteases such as Lys-C, Asp-N, or Glu-C, and the resulting proteolytic mixture are analyzed by LC-MS/MS. The peptide information from tandem mass spectrometry is used to identify peptides using various searching algorithm, and to locate modifications [35]. The bottom-up based quantification approaches can generally be broken down into two categories as shown in Fig. 3A: label-free and label-based quantification. The label-free approach is the most popular quantitative MS application for characterization and quality control of

biopharmaceuticals. This is because the goal of most MS characterization methods is to understand the primary structure and heterogeneity of relatively purified biopharmaceuticals through different stability conditions and to assess the comparability across different batches. Label-free methods often use the same MS1 peak area of interested peptides across different samples (Fig. 3B). The basis of label-free methods for quantitation is that under well controlled conditions the MS ion signal intensity from the ion source correlates with ion concentration or concentration of injected peptides. Several quantification strategies have been developed based on this principle for both relative and absolute quantification using label-free approaches (Fig. 3). For example, to assume the modified and unmodified peptides have the same MS ionization efficiencies, the label-free methods can be used to report protein modification level quantitatively in biotherapeutic products. Interestingly, Xie et al. demonstrated that N-deaminated peptides had essentially identical ionization efficiency as their unmodified counterparts and there was only a 10% MS signal difference for selected M-oxidized and unmodified peptides [36]. The label-free relative quantification for focused peptide mapping using both low- and high-resolution MS is a widespread practice for relative quantitation of protein modifications in biopharmaceutical products (Table 2). Derived from peptide mapping-based workflow, the peptide-based multi-attribute method (MAM) using high-resolution MS is an emerging tool that has been widely explored

Table 2

The properties of the list of the potential analytical methods to monitor product and process-related impurities.

Attributes	Quantitative MS or alternative assay	General principle/applications	Pros	Cons
PQA	high-resolution (e.g., Orbitrap/TOF)	high resolution accurate mass measurement of proteins or peptides	high specificity, potential replacement of methods monitoring size and charge variants	relatively higher training required for QC analyst
	low-resolution (e.g., single quadrupole) orthogonal assay (e.g., HPLC or CE based method)	low resolution and less accurate mass measurement of peptides intact molecule instead of single PQA	easy maintenance and method transfer, lower cost conventional platform assay	less specific, hard for new peak detection less QbD driven
HCP	non-targeted proteomics (DDA or DIA)	identification and relative quantification of all HCPs over certain LOD/LOQ	unbiased platform methods for global HCPs	relative low throughput
	targeted proteomics (MRM or PRM) orthogonal assay (e.g., ELISA assay)	absolute quantification of certain HCPs over certain LOQ quantification of total or single HCPs	absolute quantification high throughput, single number	standards required for calibrations, take weeks to develop methods antibody required, coverage can be low, takes years for critical reagents development

across the industry for the discovery, development, and manufacture of biopharmaceuticals [37–46].

A label-free approach was also used to estimate protein abundances with spike-in recombinant proteins without any isotope labelling. Several strategies that use peptide peak areas or peptide spectral match (PSM) (spectral counting) for quantification (Fig. 3B), such as Top3, iBAQ, and empAI, have since been published [47–50]. Those quantification approaches are reported to quantify protein abundance with reasonable accuracy over a dynamic range of 2–4 orders of magnitude. Among those label-free MS-based quantification methods, the Top3 method combining the peak areas of the top 3 peptides with the largest peak areas is widely used for relative or absolute quantification of host cell proteins in biopharmaceutical products [51,52], as the Top3 peptide peak area is directly proportional to protein abundance over the full quantification range [53]. It has been shown most process-related impurities host cell proteins (HCPs) derived from the host organism during biopharmaceutical manufacturing should be quantifiable within 2-fold of their actual value with the top 3 method [52,54,55]. Frequently, a quick relative quantification as provided by the label-free quantification will be sufficient to gain product understanding and guide subsequent process development. The obvious disadvantages are the multiple occasions for quantification error to occur during sample processing and analysis as sample recovery and LC-MS performance may vary, as well as the presence of interfering substances (detergents, abundant background

proteins). Some of these limitations can be minimized by normalizing between runs with spiked-in calibrants or by using abundant non-changing peptides as landmarks between runs (Table 2).

For protein identification and quantification, there are two main data acquisition strategies: data-dependent and data-independent acquisitions (DDA and DIA, respectively, Fig. 3A and Table 2). Unlike the traditional DDA approach with the limitation of fragmentation with low-abundance ions, with DIA, all ions within given m/z ranges are simultaneously fragmented regardless of their signal intensity [56,57]. The unbiased nature of the acquisition overcomes the limitations of DDA LC-MS/MS experiments and allows the identification of low-abundance peptides and improved reproducibility of quantification. DIA LC-MS on platforms from various vendors (e.g., LC-MS^E, SWATH) has recently widely used for biopharmaceutical product characterization [36,58–60].

Compared to label-free quantification approach, label-based quantification methods introduce isotope labels into amino acids through metabolically (e.g., stable isotope labeling by amino acids in cell culture, SILAC [61]), chemically (e.g., isobaric tags for relative and absolute quantification (iTRAQ) [62], tandem mass tags (TMT)) [63], enzymatically (O [18] labelling) [64], and synthetically (e.g., heavy isotope labelled peptides) (Fig. 3A) [65]. For relative and absolute quantification using stable isotope labeled peptides or different Isobaric reporters, one assumes equivalent ionization efficiency and MS signal response, thus, relative and absolute quantification of a sample of interest can be

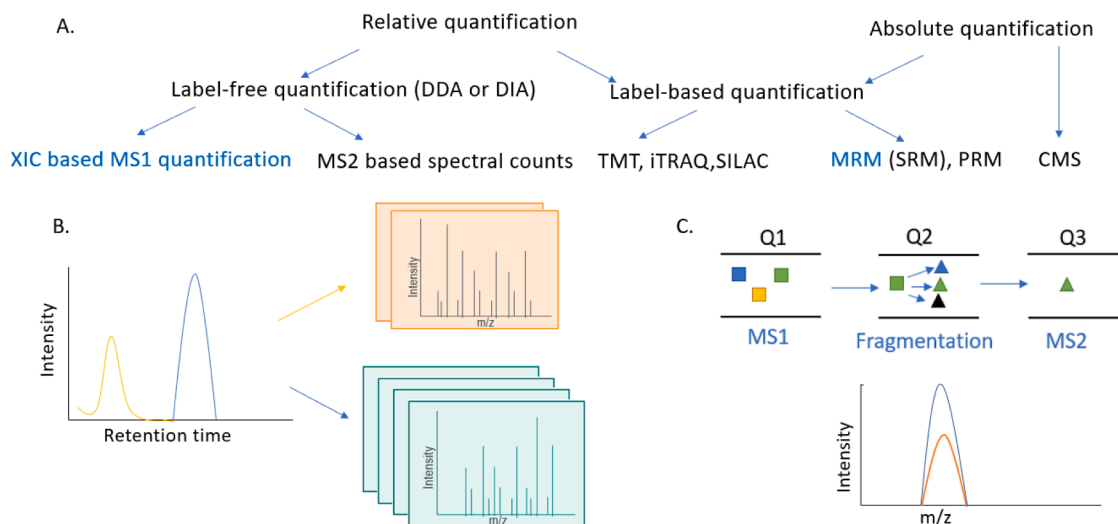


Fig. 3. Bottom-up MS-based protein quantification. A. Different protein quantification approaches including relative and absolute quantification; B. The two different protein quantification approaches based on XIC based MS1 quantification (left), and MS2 based spectral counts (right); C. The principle and typical spectrum of MRM-based absolute quantification with isotopically labeled internal standard.

performed by comparing their MS intensity in the same sample. To improve assay throughput, Mao et al. developed a TMT multiplexed quantitation approach by targeted high-resolution MS, which enables the simultaneous quantitation of mAb modifications for multiple samples in a single LC-MS run [66]. Similarly with TMT labelling, Takagi et al. simultaneously evaluated the immunoreactivity profiles of eight commercial anti-HCP antibodies with different host animals and the antigens used for immunization [67]. In general, the present utilization of labeling techniques for relative quantification in biopharmaceutical product characterization is constrained. However, we expect these methods to find broader applications in studying the growth and metabolism of engineered cells, which will inevitably influence the productivity and quality of biopharmaceuticals [68–71].

To further improve the quantitative performance, such as accuracy, sensitivity, and robustness, target MS methods such as SRM (selected reaction monitoring)/MRM (multiple reaction monitoring) or PRM (parallel reaction monitoring) have been introduced into biopharmaceutical industry for quantification of protein modifications, sequence variants, and HCPs (Fig. 3A and Table 2) [36,51,65,72–75]. Both MRM and PRM have been developed to monitor a target analyte ion and one or more product ions generated by fragmentation in the mass spectrometer (Fig. 3C). As an example, in the analysis of residual HCP during a process, synthetic peptides labeled with heavy isotopes can be added to the sample across a wide concentration range. This allows for the quantitation of HCP down to sub-parts per million (ppm) levels in routine biopharmaceutical analysis [65].

Accurate protein quantitation requires a known concentration of internal standards such as isotope-labeled or non-labelled protein/peptide standards, which could be time-consuming and costly to acquire. To eliminate the need for using standards or calibration curves, a coulometric mass spectrometric (CMS) approach for absolute protein quantitation, based on the electrochemical oxidation of oxidizable surrogate peptides combined with mass spectrometric measurement of the oxidation yield, has been developed (Fig. 3A) [76–78]. In this method using CMS, the total integrated current resulting from the electrochemical measurement is captured. As per Faraday's Law of coulometry, this measurement corresponds to the number of moles of oxidized peptide. The ion intensity ratio of the target peptide before and after oxidation provides an estimate of the fraction of the peptide that has been oxidized, from which the total amount of peptide is calculated. Presently, one of the key challenges associated with CMS is accurately quantifying proteins with low abundance levels in complex sample matrices. Efforts aimed at simplifying the sample complexity at both protein and peptide levels, as well as optimizing the CMS device, could potentially enhance the sensitivity of the assay for large-scale absolute quantification without the need for standards.

3. Quantitative MS for product and product-related variants characterization

In the production and manufacturing of biopharmaceuticals, ensuring the correct and consistent assembly of various components is of paramount importance for maintaining critical quality attributes in bi- and multi-specific antibodies, as well as ADCs. Composition integrity can often be quantitatively measured by intact or subunit LC-MS analysis. For example, Jiang et al. described the use of intact and reduced LC-MS to characterize the different forms of antibodies from a bispecific antibody using mixed mode size exclusion chromatography [79]. To quickly assess the integrity of an ADC, Firth et al. developed a quick subunit LC-MS method to calculate drug-antibody ratio (DAR) after the combination of deglycosylation, IdeS digestion, and reduction steps [80].

In contrast to antibody-based biologics, the quantification of viral structural protein abundance or their ratios holds significant importance for evaluating the potency, infectivity, and transduction efficiency of viral vector-based vaccines or gene therapy products [81,82]. For

example, vaccines were historically released using *in vivo* potency assays requiring immunization of dozens of animals. Contemporary vaccines rely on a diverse range of analytical tools, including semi-quantitative assays such as PCR, Western blot, or immunochemistry, to effectively demonstrate the transgene expression of viral vectored vaccines and assess their potency [83]. Immunoassays typically employ antibodies that are derived from animal blood serum, which can be resource-intensive and time-consuming in their production. In order to minimize the reliance on animal testing, vaccine potency testing can greatly benefit from the use of analytical tools that are both animal- and antibody-free, such as quantitative MS. The reagent-free approach utilizing LC-MS is particularly valuable for pandemic preparedness, as it provides a critical reagent-free solution. For example, Fierce et al. demonstrated a unique quantification of immunoreactive viral influenza proteins by immunoaffinity capture followed by MRM MS [84]. The quantitative MRM method is an alternative to the regulatory potency requirement using single radial immunodiffusion (SRID) assay as it has dual purpose of potency and content determination in response to a pandemic influenza threat [84]. Similarly, gene therapy utilizing adeno-associated virus (AAV) as a vector has emerged as a promising therapeutic approach with the potential to bring about significant disease modification or even cures for various monogenic disorders [85]. For non-envelope viral vaccine agents, such as AAVs, the capsid stoichiometry of the viral capsid is a CQA that can affect the tropism, infectivity, and transduction efficiency. Quantitative MS has been shown to offer distinct advantages in accurately estimating viral protein (VP) ratios in non-envelope viruses, as compared to traditional techniques such as LC or CE [81,86,87]. Recently, Wu et al. have developed a two-dimensional LC-MS method that enables simultaneous multi-attribute characterization of AAVs. This innovative approach allows for quantification of empty and full capsids, VP ratios, and protein modifications, all within a single assay [88].

In addition to biopharmaceutical product characterization, quantitative MS has found extensive application in the characterization of product-related variants, including analysis of sequence variants and protein modifications across different modalities. Biopharmaceutical products can undergo a plethora of naturally occurring or chemically induced protein modifications during their processing, manufacturing, and storage stages. Those PQAs include a variety of protein modifications, such as glycation (+162 Da), oxidation (+16 Da), deamidation (+1 Da), terminal variants and glycosylation. Extensive endeavors have been undertaken in the realm of research to comprehend the effects of PQAs on the biological potency and efficacy of biopharmaceutical products [8]. For example, deamidation on the VSNK peptide in most mAb products had a functional impact on antibody-dependent cell-mediated cytotoxicity (ADCC) and FcγRIIIa binding for IgG1 mAbs [89]. Furthermore, deamidation on the surface of AAV Capsids has been revealed to have an impact on charge heterogeneity and can alter vector function [90]. The utilization of quantitative MS methods, including intact or subunit LC-MS, as well as peptide mapping approaches, is imperative for the characterization of product-related variants and their inclusion in regulatory submission [18].

4. Quantitative MS for process-related impurities characterization

During the process development stage, it is crucial to closely monitor process-related impurities, such as HCPs and other process impurities, as they are potential critical PQAs that can impact product quality [91–93]. HCPs which are derived from the organisms utilized in the production of various biopharmaceuticals, can have an adverse impact on bioproduct efficacy and patient safety [91–93]. HCPs can have detrimental effects on biopharmaceutical products and excipients, including immunogenicity, altered biological activities, and enzymatic activities [94–97]. For example, one of the most well studied HCP phospholipase B-like 2 (PLBL2) was found to have dose-dependent immunogenicity response in

clinical trial by the detection of anti-PLBL2 antibody [98]. However, no correlation between safety events and anti-PLBL2 antibodies was observed. Additionally, no impact on the incidence of anti-drug antibodies was observed, suggesting the lack of an adjuvant effect from PLBL2 [98]. Besides the immunogenicity risk, residual HCPs can also cause serious adverse effects due to their direct biological activity as described for Chinese Hamster Ovary (CHO) monocyte chemoattractant protein-1 (MCP-1) and *E. coli* HCP Flagellin, and ATPase activity in the ChAdOx1 nCov-19 vaccine [92,99–101]. Lastly, as productivity and formulation concentration have increased, there has been a higher occurrence of cases where the stability of drug substance and formulation excipients has been impacted by HCPs. This can result in a significant impact on product shelf-life. For example, we identified the first residual HCP hexosaminidase B (HEXB) associated with mAb protein modification N-glycan degradation under accelerated and stressed stability conditions [97]. Moreover, several members of the cathepsin protein family have been reported to degrade mAb-based drug substance during stability studies [95,102–106]. Additionally, a group of proteases with lipase or esterase activities have been involved with polysorbate (PS) 80 or 20 degradation [96,107–114]. In addition to HCPs, other process impurities commonly used in bioprocessing, such as serum albumin, benzonase, trypsin, trypsin inhibitors, and low molecular weight process-related impurities, are also recognized as potential risks that need to be carefully considered during product development [115–117]. These impurities are commonly found in complex biopharmaceuticals, such as viral vaccines and gene therapy products.

To meet regulatory guidelines and to assure product quality and safety, the measurement and control of process-related impurities is mandated by regulatory agencies [92,93]. Many traditional methods have been used to characterize HCPs, such as a total protein assay (e.g., bicinchoninic acid (BCA) and Bradford assay), SDS-PAGE, or an immunoassay including ELISA [91–93,118]. ELISA has traditionally served as the primary assay used for regulatory submission and high throughput testing of total HCP during process development.^{91–93} ELISA provides a simple immunological measure of the total impurity level expressed as immuno-equivalent nanograms of HCPs per milligram drug substance by comparing to a standard. While ELISA offers advantages such as high sensitivity, throughput, ease of use, and implementation in manufacturing settings, there are recognized limitations associated with this method. These limitations include lack of specificity, inadequate antigen coverage, and non-linear dilution, which have been well acknowledged in the field (Table 2).^{91–93} A recent example of this limitation was observed in the development of a recombinant protein vaccine against SARS-CoV-2, where a polyclonal antibody reagent used to detect the SARS-CoV-2 preS protein was found to also recognize glycosylated HCPs. This resulted in a significant overestimation (approximately four to six times) of the concentration of the preS protein. As a consequence, lower antigen doses than planned were administered, and higher than anticipated levels of HCPs were present in the formulations, leading to lower immune responses than expected, particularly in older age groups, and higher reactogenicity after two doses [119]. Moreover, ELISA for total HCPs offers no information about the identity and amount of individual HCPs, which is critical for process development and risk assessment in biopharmaceutical manufacturing. The individual HCPs information provided by LC-MS-based proteomics approach ensures product quality and safety, facilitates process optimization, enables risk assessment and mitigation, ensures regulatory compliance, and supports comparability studies during process changes (Table 2).^{35, 91–93} It is not uncommon to observe that the quantification of a single HCP using LC-MS may exceed the total HCPs measured by an ELISA assay [55].

HCP characterization by LC-MS faces a significant hurdle due to the wide concentration range of low-abundance HCPs, particularly in mAb-based biologics, when compared to drug substances. Modern MS instruments typically have a dynamic range of 4–5 orders of magnitude. Notably, certain groups of high-risk HCPs, including lipases or esterases

known to be responsible for the degradation of polysorbate excipients in formulations, have been reported to exhibit activity at levels below parts per million (ppm) during the shelf life of the biologics in liquid formulations [107,114,120,121]. Various approaches have been reported to address the dynamic concentration range challenges associated with low-abundance HCPs in sample preparation, aiming to improve their identification [122], such as native digestion [123], ProteoMiner [124], drug substance depletion with Protein A [125], anti-HCP affinity chromatography [120], molecular weight cut-off [126], Hydrophilic interaction liquid chromatography (HILIC) enrichment [127], and chemical probes enrichment with activity-based protein profiling [109]. Apart from sample preparation, optimization of LC and MS methods has been undertaken to facilitate the identification and quantification of low-abundance HCPs. Techniques such as nano LC, longer single-dimensional LC, or 2D-LC have been demonstrated to enhance the detection of low-abundance HCPs during LC separation [121,128,129]. Furthermore, gas phase separation techniques utilizing ion mobility, such as high field asymmetric waveform ion mobility spectrometry (FAIMS) [125], along with diverse mass spectrometry acquisition methods, such as BoxCar [129] and DIA [55,72,130,131], have been effectively employed to enhance the identification of HCPs.

In addition to the identification of individual HCPs, non-targeted proteomics offers a means for relative quantification of HCPs (Table 2). Label-free LC-MS/MS quantification methods have been proposed, whereby all identified proteins are assigned an estimated abundance, enabling quantitative analysis of HCPs without the need for specific labeling techniques. Absolute quantification using label-free methods can be achieved by comparing HCP abundance to that of drug substances, such as the heavy or light chain of the mAb product, or spike-in recombinant proteins. This label-free quantification approach is particularly valuable for tracking HCPs across various purification steps, enabling thorough monitoring of impurity clearance and demonstrating manufacturing consistency and robustness. Obtaining accurate absolute quantification of each HCP, particularly low-abundance HCPs, through global label-free proteomics studies can be challenging due to potential differences in ionization efficiency among HCPs in MS. To achieve accurate absolute quantification, isotope dilution MS, an analytical technique based on the modification of the natural isotope composition of elements or compounds after the addition to the sample of an enriched isotope or an isotopically labeled form of the analyte, remains the gold standard for protein quantification [28]. The absolute quantification can be achieved by MRM with a triple quadrupole MS or PRM with a high-resolution MS (Table 2). The isotope dilution MS approaches have been used to quantify a few problematic HCPs [54,65,97,132,133]. For example, Gao et al. introduced a multiplex LC-MRM assay that simultaneously monitors two high-risk lipases (PLBL2 and LPLA2 (Lysosomal phospholipase A2)) known to impact biologics product quality or patient safety. The method was linear over a dynamic range of 1–500 ng/mg and showed very good correlation with ELISA assay for a single HCP [65]. The LC-MRM approach was also used to quantify another HCP HEXB associated with N-glycan degradation [97]. This targeted absolute quantification approach provided a rapid and accurate means of monitoring high risk HCP clearance for in-process support and can actively guide process improvement and optimization [65,97]. Other than process-related protein impurity, targeted absolute quantification has been used to monitor low molecular weight process-related impurities for process development. For example, Jiang et al. developed and validated a LC-MRM-MS method for quantification of protease inhibitor Pepstatin A to monitor its robust clearance in a recombinant protein-based vaccine downstream process [117]. This method is currently being applied to routine process development experiments and site-to-site comparability studies [117]. In addition to isotopically labelled absolute quantification, standard-free absolute quantitation by coulometric MS was recently applied for the measurement of 500 ppm of PLBL2 in the presence of a highly abundant mAb [78].

Despite the great advantages of LC-MS methods over ELISA for

identification and measurement of HCPs, perceptions regarding method throughput, reproducibility and robustness have to date limited the adoption of LC-MS platforms as the primary method for HCP characterization. Recent advances in high-throughput proteomics have increased sample testing speed and reproducibility. Those advances include 1) high-throughput, automatic sample preparation using liquid-handling workstations with 96- or 384-channel microplate formats; [134] 2) rapid LC systems (e.g., Evosep ONE) [135,136] or micro-flow LC; [137] 3) state-of-the-art MS instrumentation with high sensitivity and fast scanning speed to achieve in-depth proteomics. In particular, ion mobility-based MS, such as TimsTOF Pro [138] or Exploris 480 equipped with a FAIMS system; [139] and 4) new algorithms and machine learning-based proteomics data analysis software packages [140–142]. Deep machine learning methods, such as convolutional neural networks (CNNs) and recurrent neural networks (RNNs), have been increasingly applied to achieve higher sensitivity and/or specificity in proteomics data analysis [143]. Recently, Derks et al. developed an experimental and computational framework, called plexDIA, for simultaneously multiplexing the analysis of peptides and samples for high-throughput of sensitive proteomics. Using 1-hour active gradients, plexDIA quantified ~8000 proteins in each sample of labeled three-plex sets [144]. To improve the throughput of HCP testing, Ma et al. developed a sensitive, robust, and reproducible workflow for HCP detection and quantification in a significantly shorter turnaround time than that previously reported using an Evosep ONE LC system coupled to an Orbitrap Fusion Lumos mass spectrometer [135]. Because of its fast turnaround time, this HCP workflow can be integrated into process development for the high-throughput (60 samples analyzed per day) identification of HCPs [135]. In addition to taking into account throughput, LC-MS based proteomics workflow that is properly optimized often exhibits excellent reproducibility. For example, Husson et al. evaluated 3 different HCP quantification approaches, which displayed equivalent and good precision with a vast majority of CV (Coefficient Variation) values between technical triplicates below 5% [55]. The reproducibility and robustness of the entire LC-MS workflow can be greatly enhanced from those standardized procedures on sample preparation, LC-MS, and data analysis.

5. Quantitative MS in GMP/QC for manufacturing

With the growing molecular complexity of biopharmaceuticals and the ever-evolving regulatory requirements, there is a growing need to integrate MS into GMP manufacturing processes. This transition is aimed at achieving improved product understanding and increased testing efficiency [145,146]. Two distinct workflows utilizing MS have

emerged and are being utilized in GMP and Quality Control (QC) for the release and stability testing of biopharmaceuticals [145,146]. One approach emphasizes intact or subunit-based MS analysis, while the other approach centers around peptide mapping-based analysis. Table 3 provides an overview of the ATP and business motivations driving each approach, in addition to conventional assays employed to monitor impurities related to product and process.

Due to its straightforward sample preparation, reliable relative quantification, and efficient data analysis, intact/subunit-based quantitative MS has gained popularity as the preferred method for GMP applications (Table 3). Habegger et al. reported the successful application of MS analysis of reduced antibody for quantitative monitoring of the heavy chain Fc galactosylation of trastuzumab as part of the drug substance release testing for over a decade [147]. Since 2004, over 500 batches of drug substance have been successfully released using MS analysis for QC testing of Trastuzumab. Remarkably, only three system suitability deviations have been observed for this subunit MS method, indicating its robustness and suitability for QC testing in a GMP environment. This highlights the wide applicability of the subunit MS method and its consistent performance in ensuring product quality [147]. Recently, Sokolowska et al. described the implementation of an automated, high-throughput, and GMP compliant subunit LC-MS method for quantifying antibody Fc methionine oxidation for commercial product release and stability testing [148]. This subunit-based MS method was co-validated across three laboratories including one commercial QC lab using GMP compliant MS hardware and software. This study highlighted the advantage of applying the subunit MS method to quantify oxidation profiles, and the potential for monitoring additional PQAs, such as glycosylation, glycation, sequence variant, and terminal modifications [148].

To address those challenges of monitoring low-level PQA (e.g., <1%), or PQA with small mass change, peptide mapping-based MS analyses provides a more attractive option. Despite increased complexity with sample preparation and data analysis, peptide mapping based quantitative MS approaches, such as MAM, have recently gained significant interest [46,149,150]. Peptide-based quantitative MS workflows often use a more complicated sample preparation procedure which requires special attention to achieve robust protein digestion, while minimizing artifactual modifications, such as deamidation, isomerization, and oxidation. To address this, Ren et al. developed a fast and reliable trypsin digest approach to minimize digestion artifacts using desalting column [151]. In order to achieve optimal digestion with the aid of automated robotics, specialized desalting plates or tips are frequently utilized to streamline the process [40,152–157]. To eliminate the need for desalting steps and simplify sample preparation procedures,

Table 3

The ATP and business drive for each analytical method to monitor product- and process-related impurities.

Assay	attributes	PQA				HCP centralized		
		Subunit MS	Focused peptide mapping	Peptide mapping-based MAM	HPLC/CE	Untargeted	Targeted	ELISA
		PQA	PQA	PQA/NPD	intact molecule	global HCP	individual HCP	total or individual HCP
Assay Performance	accuracy	7	7	10	4	4	10	4
	precision	7	7	7	10	4	7	10
	LOD/LOQ	4	4	10	1	4	10	10
	specificity	4	4	10	1	10	7	1
	linearity	4	7	7	4	4	10	7
Business drivers	range	4	4	10	4	4	7	10
	throughput	7	4	1	10	1	7	10
	turnaround time	7	4	1	10	4	7	10
	critical reagents	10	10	10	10	10	4	1
	equipment	4	10	4	10	4	7	10
	GMP readiness	7	7	4	10	4	7	10

For each ATP requirement, a score is given as follows: “1”, does not meet; “4”, partially meets; “7”, meets; and “10”, exceeds the requirements.

we have developed a rapid enzymatic digestion approach utilizing Lys-C. This innovative method greatly facilitates high-throughput and automated digestion, making it more accessible and efficient for implementation [158]. On the MS side, both low- and high-resolution MS have been used for peptide mapping-based quantitative MS in GMPs (Table 3). When it comes to focused peptide mapping for GMP and QC purposes, low-resolution mass spectrometers, such as single quadrupole mass spectrometers, offer several advantages over high-resolution MS. These include their ease of use, as well as the convenience of a single software platform for both LC and MS, simplifying the analytical workflow. To monitor product PQA and use as an identity test, Evans et al. developed and validated a targeted multi-attribute monitoring method using the low-resolution MS system with a fully automated data processing workflow that is suitable for identity testing, sequence variant control, and CQA quantitation in commercial QC labs [39]. Similar focused peptide mapping approaches have been established and qualified in several other labs for PQA monitoring [43,45,159]. In regulated GMP and QC laboratories, focused peptide mapping can serve as a valuable tool for targeted assessment of specific sites of modification, particularly in cases where CE or LC based methods may not be practical or feasible. Like the subunit-based MS approach, low-resolution MS-based peptide mapping may face challenges in accurately tracking PTMs with small mass changes, such as deamidation, especially in the absence of distinct chromatographic separation (Table 2).

To improve testing efficiency and specificity for tracking those challenging PQAs and ultimately replace certain LC or CE based traditional assays, high-resolution MS based peptide mapping coupled with new peak detection (NPD), or MAM, offers the most advantages over intact/subunit-based MS or low-resolution MS-based focused peptide mapping approaches [46,149,150]. A typical peptide mapping-based MAM workflow is shown in Fig. 4. Proteins are digested into peptides using appropriate enzymes (e.g., trypsin or Lys-C), which are subsequently analyzed using high-resolution LC-MS with MS/MS fragmentation for detailed product characterization. For routine MAM applications using GMP compliant high-resolution MS instrument and software, the PQAs extracted from the characterization workflow are monitored with a targeted data analysis workflow, and new peaks compared to a reference standard are detected. In 2015, Rogers et al. described for the first time the use of high-resolution MS for monitoring multiple product attributes for product characterization and process development [42]. This study lays out the promise that MAM can potentially replace or reduce the number of assays needed for process development, product disposition, and in-process control [42,46,149]. Following the groundbreaking study, numerous companies have made

significant progress in developing or qualifying high-resolution MS-based peptide mapping methods for the characterization and quality control of biopharmaceuticals [40,152–154,156,158,160–163].

Although MS has not yet been widely used in GMP environment as a QC test for biopharmaceuticals, the steps for implementation of MS in GMP testing for release or stability testing are similar as any QC analytical methods. Phase appropriate method qualification and validation of MS assays is critical for its use in GMP [146,149]. As experiment variables listed in Fig. 5 and Table 4, this validation would assess the assay performance and robustness of the entire workflow of a method including sample preparation, LC, MS, and software capabilities. The overall validation strategy often followed the ICH Guideline Validation of Analytical Procedures: Text and Methodology Q2 (R1) [41,149]. Depending on the circumstances under which the MS assay is intended to be used (Identification, Quantitation or Limit), analytical method validation of a quantitative method for PQA monitoring would include assessments for specificity, linearity, accuracy, precision, range, and quantitation limit as shown in Table 5 [41]. Due to the complexity of MS assays as compared to other traditional LC-based methods and the lack of standards for certain PQAs, the approaches to evaluate these validation characteristics may not be entirely straightforward, especially for linearity, accuracy and limit of quantification [41]. Thus, a phased approach for method qualification and validation of MS assays is critical to understand the MS assay performance. We evaluated the assay performances of a platform MAM workflow for identity, PQA monitoring and NPD using an IgG4 mAb-1 and its cofomulations as model molecules [41]. Samples with different protein load and various levels of protein modifications were used to evaluate the analytical assay performance of MAM. The assay performance evaluation demonstrated the full potential of the platform MAM approach for its intended use for characterization and quality control of a single mAb and a mAb in its cofomulations [41]. The qualification/validation approaches used for various MS based assay and their analytical performances are summarized in Table 5. Overall, the analytical performances are considered well acceptable for their potential GMP use. Once method validation is successfully completed and the method is deemed suitable, full method robustness evaluation can be performed. Robustness evaluation assesses the method's ability to remain unaffected by small, deliberate variations in operating parameters as shown in Fig. 5 and Table 4. Design of experiments (DOE) design is often essential for determining the robustness of an analytical assay by enabling a systematic evaluation of factors and their interactions, which helps identify critical method parameters and assesses the method's tolerance to minor variations that may be encountered during routine use (Table 4) [45,148].

Beyond method validation, the successful implementation of MS

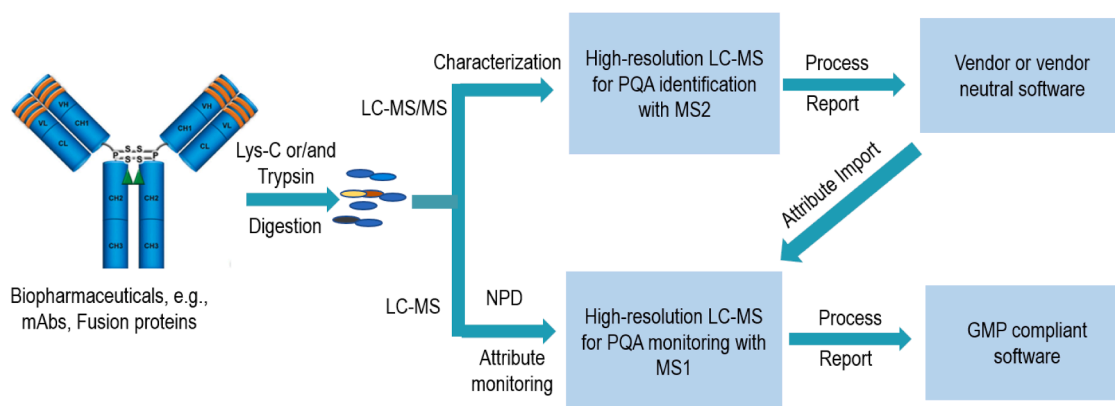


Fig. 4. A typical peptide mapping-based MAM workflow. Proteins are digested into peptides using an appropriate enzyme (e.g., trypsin or Lys-C), which are subsequently analyzed using high-resolution LC-MS with MS/MS fragmentation for detailed product characterization. For routine MAM applications using GMP compliant high-resolution MS instrument and software, the PQAs extracted from the characterization workflow are monitored with a targeted data analysis workflow, and new peaks compared to a reference standard are detected.

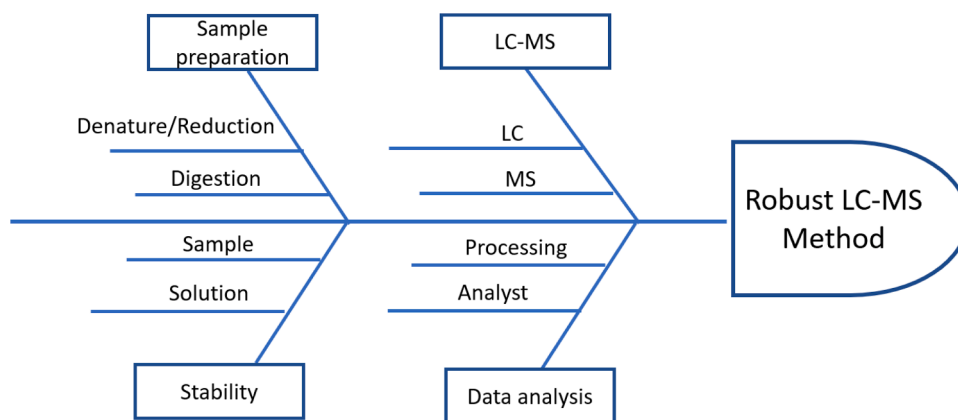


Fig. 5. The Ishikawa or Fishbone diagram to identify potential variables for a robust LC-MS method for biopharmaceutical development. The individual assay variables for each major categories (sample preparation, LC-MS, data analysis and sample/solution stability) are summarized in Table 4.

Table 4

The experiment variables to be considered for a robust quantitative LC-MS assay.

Assay components	Assay variables*	Selected references
sample preparation	reducing/alkylation concentration, temperature and time, digest buffer pH, enzyme concentration and digestion time	[45,148,154, 158]
LC	initial % mobile phase B [45], % mobile phase additives, protein load [148], column temperature [45], flow rate	
MS	probe temperature, capillary temperature/voltage, sheath/aux gas flow rate, Aux gas temperature, cone voltage, desolvation temperature, MS1 AGC target, collision energy	
data analysis	peak integration, analyst	
sample/solution stability	protein freeze/thaw, digested peptides freeze/thaw, autosampler stability [158], mobile phase solutions	
equipment and material	column lots, LC, MS, lots for reducing/alkylation reagents and enzyme, vial	

* Assay variables in bold may have major impacts in assay performance for certain PQAs (reference included)

assays, such as MAM as a primary GMP assays to replace conventional assays requires a comprehensive risk assessment, comparison to conventional assays, and NPD evaluation [149]. For risk assessment of MAM as the primary assay for release, the likelihood and risk of MAM to successfully monitor each individual CQA of the target molecule, and a comparison of key risk profile differences to conventional methods could be interesting topics to be included for consideration. One common risk of peptide mapping-based MAM is the complete sequence of specific proteins that are lost after enzyme digestion. This bottom-up approach contrasts with some of the conventional methods that measure the whole or parts of the molecule. For example, as monoclonal antibodies have two identical heavy chains, MAM would not be able to differentiate mAbs with modifications predominantly on one chain versus those containing modifications on both chains [149]. Furthermore, it is worth noting that when utilizing peptide mapping-based MAM, there is a possibility of overlooking significant quantities of protein fragments or clips. In such cases, NPD are expected to be capable of capturing these fragments or clips, if present. However, ensuring that the resulting peptides after clipping and enzyme digestion are of the desired size (e.g., > 5 amino acids) for accurate measurement by MS may pose challenges. As a result, incorporating at least one complementary method based on intact protein analysis, such as size exclusion chromatography (SEC), in the QC assay panel may be necessary to ensure product quality.

In order to justify that MAM can be used to replace conventional assays, the comparison with conventional assay is critical. As described previously, several conventional methods have been used to characterize and quality control of biotherapeutics during product development and release, such as HILIC for glycan profiling, Ion exchange chromatography (IEX) for charge variant analysis, hydrophobic interaction chromatography (HIC) for oxidation, and capillary electrophoresis sodium dodecyl sulfate (CE-SDS) for clipped variant analysis. Several studies have shown encouraging data that MAM can provide very comparable results compared to conventional methods [41,42, 161]. For example, we have shown that a mAb oxidation from MAM correlated very well with a conventional HIC assay. Using accelerated stability samples, Guan et al. showed excellent correlation between MAM and conventional methods including CEX-HPLC for charge variants and rCE-SDS for size variants [161]. In general, it is not surprising that the absolute values obtained from MAM and conventional approaches would differ. However, the most crucial aspect for establishing a method bridge between MAM and traditional assays is the trend and correlation between the two methods.

Detecting new peaks beyond the targeted monitoring assay is widely recognized as a crucial step to address potential regulatory objections when replacing conventional assays, as it is considered indispensable in ensuring product quality. Given the complexity of the entire workflow and variability of background ions within LC-MS-based peptide mapping, the discussion of properly setting the parameters for meaningful detection of new peaks in a peptide-based MAM assay with the acceptable false positive and false negative rates remains an unresolved element of peptide mapping-based MAM discussions [41,155]. A streamlined approach to monitoring peptides through efficient workflows is crucial for quickly detecting new peaks using precursor MS ions. This would necessitate either accessing historical databases from product developability studies or acquiring new independent peptide fragmentation data for peak identification. In the context of batch failure investigations, an integrated workflow for monitoring and characterizing new peaks is essential to ensure data integrity of the monitoring assay and prevent delays in product release due to reanalysis. While meeting the requirements of process and stability functions, the appropriateness of NPD for the QC release function remains a key discussion point and presents significant challenges.

Peptide mapping-based quantitative MS approach, such as MAM, intends to characterize and quantify most PQAs. Some PQAs, such as low abundance HCPs (<100 ppm) may not be always detected by MAM. For mAb based biotherapeutics, LC-MS is often routinely used to quantify or monitor PQAs with protein modifications levels greater than 0.1%. A product variant with 0.1% can be translated to 1000 ppm compared to the main product without modification. Not surprisingly, there are technical challenges to monitor and quantify low abundance HCPs in a

Table 5
Selected LC-MS-based method qualification/validation approaches.

Assay	Attributes	Instrument	Sample preparation	Qualification approach	Accuracy (%)	Precision (repeatability, %CV)	Precision (intermediate precision, %CV)	Specificity	LOQ (%)	Linearity	Range	Reference
subunit mass	two truncated fragments	TOF, BioAccord	Fc fusion protein, PNGase F treatment before DTT reduction	co-validation from 3 labs using different protein load	78.6–120.0	< 13.1	< 19.2	no interference from formulation blank	0.54	> =0.96	0.54% (LLOQ) 25.78% (HLOQ)	[188]
subunit mass	oxidation	Q-ToF, Xevo G2-XS	single mAb, IdeS/EndoS before DTT reduction	co-validation from 3 labs using chemically oxidized mAb1 into unstressed DS	80.5 – 117.8	< =3.6	< =11.8	no interfering peaks	1.3	0.99	1.3–54.4% of oxidation level	[148]
subunit mass	mannose-5	Orbitrap, Exactive plus	single mAb, ProA purified from cell culture before DTT reduction	single lab using mixture sample with low and elevated relative mannose-5 levels	96–105	< =11.3	< 7.8	no interfering peaks	0.8	> 0.99	0.8–11.0% of mannose-5	[189]
focused peptide mapping	deamidation	Quadrupole, QDa	co-formulation mAb, Trypsin digestion without desalting	single lab using different stressed samples	97–107	< =2.5	< =4.5	no interfering peaks	4.5	> 0.99	6.4–58.5% of deamidation level	[159]
focused peptide mapping	deamidation and succinimide	Quadrupole, QDa	single mAb, Lys-C and Trypsin digestion without desalting	single lab using different stressed samples	99–103	< =1.8	< =10.2	no interfering peaks	1.1	> 0.99	2.6–10.9% of deamidation, 4.7–33.3% of succinimide	[190]
focused peptide mapping	sequence variant, isomerization, oxidation	Quadrupole, QDa	bispecific antibody, trypsin digestion after buffer exchange	co-validation from 6 labs using different stressed samples or synthetic peptide for sequence variant	89.5–109.3	< =16.6	< =13.2	no interfering species (>0.2%)	0.6–1.0	> 0.99	CQA dependent, 0.6–21.5%	[39]
focused peptide mapping	oxidation on methionine and tryptophan	Quadrupole, QDa	co-formulation mAb, Lys-C digestion without desalting	single lab using different protein load	94–126	< =9	< =15	no peaks> =QL	0.3	> =0.98	30–125% of assay level	[45]
peptide mapping-based MAM	oxidation, deamidation, clipping, and glycosylation	Orbitrap, Q Exactive plus	Fc-fusion protein, trypsin digestion after buffer exchange	single lab using different protein load	accuracy is inferred if specificity and precision met	SD < 1	SD< =1	no significant interference	0.4–4.0	> =0.96	1–3 µg Protein Load	[161]
peptide mapping-based MAM	oxidation, isomerization, deamidation, sequence variant, and glycosylation	Orbitrap, Q Exactive plus	mAbs, trypsin digestion after buffer exchange	single lab using different lots or stressed samples	97–112	< 5	< 7	interference < 0.1%	0.02–2.4	1	0.6–79% protein modifications	[154]
peptide mapping-based MAM	oxidation, deamidation, N and C terminal variants	Orbitrap, (Q) Exactive plus	single mAb, Lys-C digestion without desalting	single lab using different protein loads and stressed samples for oxidation	81.63–122.45	< =12	< =20	up to 1.4% carry-over for one peptide	0.01–0.48	> 0.99	25–200% of assay level	[41]

typical peptide mapping workflow. For example, Jakes et al. utilized NPD in a MAM workflow to effectively detect problematic HCPs at levels as low as 100 ppm. However, the study did not provide information about the total number of new peaks detected, including those from sources other than the two identified HCPs [157]. In general, it is considered as a significant concern for certain HCPs to reach or be closer to 1000 ppm or even 100 ppm in a final drug substance. As discussed in the process related impurities section, due to the dynamic range of MS, specific sample preparation and LC-MS setup may be needed for HCPs to be reliably detected at sub-ppm level. Moving a nontargeted MS-based proteomics workflow for HCP into GMP for QC release can be challenging due to the complexity of the sample preparation, LC-MS and data analysis, and the concerns of reproducibility and robustness of the entire workflow (Table 3). It is promising to note that there has been significant interest in adopting proteomics workflows for regulated use, as evidenced by the growing number of conference presentations, roundtable discussions, and publications on method development, method qualification or validation from academia, biopharmaceutical companies, and contract research organizations (CROs) [164,165]. For example, Ghosh et al. validated a LC-MS/MS based antigen identification workflow suitable for pharmaceutical therapies according to the current FDA (Food and Drug Administration) and EMA (European Medicines Agency) guidelines [165]. The applications of MS in GMP for HCP characterization are especially useful when an uncommon cell line is used to product biopharmaceuticals, or a program is under time pressure to develop a high-quality process or product-specific ELISA kits. If one or a few HCPs have been consistently identified during early-stage process development, it is more practical to use a targeted MS workflow monitoring HCPs clearance in regulated labs since triple quadrupole-based MS have been widely used in the industry supporting regulated bioanalysis in GLP (Good Laboratory Practice) labs. The antibody-free targeted MS has advantages over ELISA on specificity, multiplex capability, and shorter method development time for single HCP monitoring (Table 3).

Companies need to take extreme cautions when evaluating MS platforms to deploy for GMP testing. The initial investment for MS platforms may be higher compared to traditional HPLC, CE or plate reader-based assays. The costs are further increased due to the need for redundancy systems, as LC-MS systems are less robust and batch release can be significantly delayed because of instrument downtime without backups. Additionally, the fast evolution of LC-MS platforms and software raises concerns about instrument or informatics lifecycle management, necessitating a long-term commitment from instrument vendors for the successful use of MS in GMP. These challenges may pose difficulties for smaller companies attempting to implement MS in GMP, which, in turn, may raise the bar for biosimilar development if MS is used for release of innovative biopharmaceuticals. Encouragingly, several CROs have started to build the capability of MS in GMP for external collaborations, which offers the alternative approach for GMP testing if desired.

Collectively, the MS in GMP paradigm for large molecules represents a visionary approach that highlights the substantial analytical and business value it can bring to biopharmaceutical development and commercialization in the long run. To achieve this vision, it will be crucial to foster effective partnerships among instrument/software vendors and regulatory agencies. Early engagement and collaboration with regulatory agencies can provide sponsor companies with a vital mechanism to proactively prepare a comprehensive data package for approval. Furthermore, ongoing discussions and knowledge sharing across the industry will be essential to accelerate the implementation of MS in GMP and shape the future of biopharmaceutical analytics.

6. Quantitative MS for process analytical technology

Process Analytical Technology (PAT) is a system for designing, analyzing, and controlling pharmaceutical manufacturing processes

through measurements of critical quality and performance attributes of raw and processed materials to ensure final product quality [166–168]. PAT plays a crucial role in optimizing biopharmaceutical processes and ensuring the consistent measurement of CQAs at various stages of manufacturing and during QC release testing. PAT is widely applied in the biopharmaceutical industry for analysis of raw materials, in-process monitoring, and final product analysis [166–169]. As PAT solutions continue to advance, they have the potential to enable real-time release testing (RTRT) of biopharmaceuticals in the future [170]. In order to fully grasp the progress of PAT in biopharmaceutical manufacturing, it's crucial to understand the diverse types of PAT operations employed. These include in-line analysis, where analyzers are placed within process vessels or streams to conduct real-time, in-situ analysis. On-line analyses, on the other hand, are connected to process side streams and perform periodic automatic sampling, with samples potentially returning to the process after analysis. At-line analyses involve collecting samples from the process and analyzing them nearby on the same process floor, as opposed to traditional off-line analysis that takes place remotely in a dedicated analytical lab. While in-line and on-line analyses allow for continuous process measurement, feedback, and control, at-line and off-line measurements introduce delays between sampling and result availability, thereby limiting their suitability for direct process control [166–168].

Seamless integration with existing process PAT systems is essential to align with the QbD concept. The potential for implementing PAT is vast and extends to every facet of biopharmaceutical manufacturing processes [166–168,171,172]. For example, in-line sensors have been used to measure pH, temperature, dissolved oxygen, and carbon dioxide [169]. In-line optical sensors (e.g., Raman, near infrared spectroscopy (NIR)) are coupled with multivariate modeling data analysis to measure certain nutrition and metabolites [173,174]. In addition, a variety of on-line tools are widely used to measure cell density (e.g., dielectric spectroscopy, fluorescence spectroscopy) [169], protein aggregates and product concentration (e.g., ProA-SEC and index of refraction) [175, 176], molecular weight (e.g., multi-angle static light scattering (MALS)) [177], and protein charge variants (e.g., IEX) [178].

Quantitative MS is recognized as a vital tool for PAT characterization of biopharmaceuticals. The diverse MS approaches potentially utilized in GMP for drug substance characterization can be adapted for potential PAT applications, serving as at-line or off-line tools for in-process control. To enable the on-line use of LC-MS for PAT, high-throughput sample preparation, fast LC-MS and automated data process are critical for feedback and control of a process without any hold times. For example, high throughput intact/reduced mass has been used to monitor glycosylation profile and major product variants, such as oxidation [148,179]. To reduce the delay and enhance the ability to achieve quality targets during process development, Dong and coworkers developed a low-volume, high-throughput, and high-content analytical platform for at-line product quality analysis, which includes an automated, 96-well plate protein A purification step to isolate antibody product from the cell culture fermentation collection, followed by rapid subunit MS analysis to quantify glycosylated and glycated species in a series of small scale experiments [179]. Protein digestion is often the bottleneck for high-throughput LC-MS analysis. Most recently, Gunawardena et al. developed a microdroplet protein digestion technique with automated sample flow injection and online MS analysis for a rapid subunit characterization of antibodies [180]. The total analysis time of this integrated approach is 2 min per sample with great assay precision (~2% coefficient of variation) [180]. Not only for subunit analysis, the microdroplet digestion approach holds great potential for ultrafast peptide mapping analysis as well as it showed similar digestion efficiency compared to the conventional digestion approach [181]. Due to the technological improvement of multi-dimensional LC-MS and on-column enzyme digestion, multiple PQAs of protein-based therapeutics were tested by unified and integrated intact, subunit, and peptide mapping approaches [182–187]. Cross-lab study demonstrated the

great reproducibility of those complicated instrument setup [187]. When coupled with automatic aseptic sampling, multi-dimensional LC-MS offers real-time online monitoring PQAs at subunit or peptide level at the manufacturing floor [185]. Dahotre and coworkers demonstrated an automated sampling system and multi-dimensional LC-MS/MS technology to quantify PQAs from a standard mAb and fusion protein during the cell culture process [185,186]. Similarly, Yang et al. demonstrated a fully integrated peptide-mapping based platform for automated real time online process monitoring [162]. The advances in aseptic sampling, automated sample preparation, LC-MS, and data analysis pave the way for real time process monitoring, feedback control for improving the process yield as well as the product quality. MS instruments, however, have a large footprint and can be expensive in general, personnel training is extensive and speed to data needs to be improved for providing timely information to manufacturing personnel during production runs in real-time [145]. Understanding the analytical performance and the robustness of the entire LC-MS workflow will enable the transition of PAT from development to potential RTRT in GMP in the future (Fig. 5 and Tables 3 and 4).

7. Conclusions and outlook

Quantitative MS has emerged as a versatile and powerful technique for various facets of biopharmaceutical characterization and quality control at different stages of development. It has proven to be particularly valuable in identifying and quantifying structural integrity, characterizing complex protein modifications, and detecting process-related impurities such as HCP contaminants in biopharmaceutical products. With advancements in sample preparation techniques, instrumentation, and data analysis methods, it is expected that the capabilities of MS analyses in biopharmaceutical quality control will continue to expand in the lifecycle of biopharmaceutical product development. The continued advancement of quantitative MS in biopharmaceutical quality control has the potential to revolutionize the field by enabling real-time, data-driven decision-making, and process optimization. This can lead to improved process efficiency and enhanced product quality, and potentially results in reduced production costs for broader global availability of innovative medicines.

CRediT authorship contribution statement

Xuanwen Li: Conceptualization, Writing – original draft preparation, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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